ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MECHANICAL TECHNOLOGY

THEORY PAPER

MARKS: 100 **TIME: 2 hours**

Topics: safety, tools and measuring, materials, joining methods, maintenance and forces. Suitable for Automotive, Fitting and Machining, and Welding and Metalwork learners.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show formulae, substitutions and units in ALL calculations. Use π = 3,14.

QUESTION 1: MULTIPLE CHOICE

1.1 Which of the following is a non-ferrous metal?

A Cast iron B Mild steel
C Copper D Wrought iron

1.2 The tool used to mark lines on metal is a …

A scriber. B file.
C hacksaw. D punch.

1.3 The property that allows a metal to be drawn into a wire is …

A ductility. B hardness.
C brittleness. D elasticity.

1.4 In gas welding, oxygen is mixed with …

A nitrogen. B acetylene.
C helium. D carbon dioxide.

1.5 An external thread (on a bolt) is cut with a …

A tap. B die.
C reamer. D drill.

(5 × 2)
[10]

QUESTION 2: SAFETY

2.1 Give FOUR general safety rules for a mechanical workshop. (4)

2.2 Give THREE safety precautions when using a bench grinder. (3)

2.3 Give TWO dangers of gas welding and a precaution for each. (4)

2.4 Which type of fire extinguisher should be used on an electrical fire? (2)

2.5 Why must oil spills on the workshop floor be cleaned up immediately? (2)

[15]

QUESTION 3: TOOLS AND MEASURING

3.1 State the use of: centre punch, try square, vernier caliper, micrometer, hacksaw. (5)

3.2 On a vernier caliper with an accuracy of 0,1 mm, the main scale reads 24 mm and the 7th vernier line lines up with a main scale line. Write down the reading. (3)

3.3 Explain how a hacksaw blade must be fitted, and what kind of blade to use for cutting thin sheet metal. (3)

3.4 Name THREE types of files by their cross-section shape. (3)

3.5 Give THREE ways to look after precision measuring instruments. (3)

3.6 Calculate the spindle speed (in r/min) for a 10 mm drill if the cutting speed is 30 m/min. (3)

Use $N = \frac{v}{\pi D}$, with D in metres.

[20]

QUESTION 4: MATERIALS

4.1 Explain the difference between ferrous and non-ferrous metals, with ONE example of each. (4)

4.2 Define the following properties of materials: hardness, toughness, malleability. (6)

4.3 State the purpose of: (a) hardening; (b) annealing. (4)

4.4 Name the main metals in brass and in bronze. (4)

4.5 Give TWO reasons why aluminium alloys are used to build aircraft. (2)

[20]

QUESTION 5: JOINING METHODS

5.1 Distinguish between permanent and temporary joining methods. Give ONE example of each. (4)

5.2 Name THREE items of protective equipment needed for arc welding. (3)

5.3 Give ONE situation where rivets are used instead of welding, and explain why. (2)

5.4 What is the function of a washer under a nut? (2)

5.5 Explain the difference between soldering and welding. (3)

5.6 Define the following screw-thread terms: pitch, crest, root. (3)

5.7 An M10 × 1,5 bolt has a pitch of 1,5 mm. How many complete threads are there along 30 mm of the bolt? (3)

[20]

QUESTION 6: MAINTENANCE AND FORCES

6.1 What is preventive maintenance? Give TWO benefits. (4)

6.2 Give TWO purposes of lubrication in machines. (2)

6.3 A force of 80 N is applied at the end of a spanner 0,25 m long. Calculate the turning moment (torque). (3)

6.4 Give an example of a part that experiences each type of force: tension, compression, shear. (3)

6.5 In a belt drive, the driver pulley has a diameter of 100 mm and turns at 1 400 r/min. The driven pulley has a diameter of 200 mm. Calculate the speed of the driven pulley. (3)

[15]

TOTAL: 100

END OF QUESTION PAPER